

Imperial College London
Department of Earth Science and Engineering
MSc in Applied Computational Science and Engineering

Independent Research Project
Project Plan

Context to Consequence - AI-Driven News Categorisation and Event Prediction

by
Shuhan Liu

repository: <https://github.com/ese-ada-lovelace-2025/irp-sl1425.git>

Supervisors:
Antony Sommerfeld

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Abstract

Some early text-based election forecasting studies used sentiment analysis, but this approach may suffer from population bias and limited local representation. This project proposes a context-to-consequence framework that uses real political news to support constituency-level voting outcome prediction. News articles will first be filtered and processed by an LLM into structured context representations. These representations will then be used to explore consequence-aware embeddings through contrastive learning. The embeddings will be combined with available constituency-level data for downstream vote prediction. If feasible, the project may also include baseline comparisons and a constrained reflection agent for error analysis. The expected outcome is a prototype pipeline that connects political news context with voting outcome prediction.

1 Problem Description and Significance

In UK general elections, Members of Parliament are elected under the First-Past-the-Post (FPTP) system, where each constituency elects the candidate with the most votes [2]. Therefore, constituency-level prediction is an important challenge in British election forecasting.

Current constituency-level forecasts often rely on opinion polls and constituency-level population information [5]. A common approach is Multilevel Regression and Post-stratification (MRP), which estimates small-area public opinion from large national samples [4]. For example, Ipsos used MRP to project individual seat outcomes in the 2024 UK General Election [5].

However, these approaches mainly rely on polling data and relatively stable population information, rather than directly modelling political news during the campaign. This project therefore explores how real political news can be converted into structured context representations and combined with constituency-level data for voting outcome prediction.

2 Existing Work

Early studies have used social media text and sentiment analysis for election forecasting. Burnap et al. used Twitter data and automated sentiment analysis to forecast the 2015 UK General Election, using the 2010 results as a baseline for constituency-level seat prediction [1]. However, their study also shows limitations of Twitter-based forecasting, including population bias and difficulty capturing geographically concentrated party support in a multi-party system [1]. This suggests that simple sentiment signals are not enough to model the wider context and constituency-level consequences of political information.

Beyond social media sentiment analysis, recent work has used LLMs for political decision-making simulation. Yu et al. propose a multi-step reasoning framework that uses demographic, temporal, and ideological factors to simulate voter decisions [8]. However, their study also shows that LLM-based political simulations may contain political bias and may over-rely on demographic voting patterns [8]. They also note that their method does not fully capture dynamic factors such as changing public opinion, media narratives, or unexpected events [8]. This shows a limitation of simulation-based approaches and motivates this project to focus instead on real political news and its possible relationship with constituency-level voting outcomes.

Wang et al. provide an important reference for news-based forecasting by proposing an LLM-based framework that combines news events with time series data [7]. A key finding is that news should be filtered before being used for prediction, because unfiltered news may introduce noise and incorrect causal information [7]. Their framework uses LLM-based reasoning agents to select relevant news, consider short-term and long-term effects, and explain why selected events may affect the target variable. It also uses an evaluation agent to check missed news

based on prediction errors and update the selection logic [7]. This is useful for this project because it shows that news should be connected to the prediction target through relevance filtering and contextual reasoning. While Wang et al. apply this idea to numerical time series forecasting, this project adapts it to political news by exploring how news context may relate to constituency-level voting consequences.

To further connect political news context with voting outcomes, this project also draws on contrastive learning to connect political news context with voting outcomes. Khosla et al.'s Supervised Contrastive Learning uses label information to pull samples from the same class closer together and push samples from different classes further apart [6]. Gao et al.'s SimCSE shows that contrastive learning can also be used to learn effective sentence embeddings [3]. Building on these ideas, this project will explore whether news with similar possible voting consequences can be represented closer together, while news with different consequences can be more clearly separated.

3 Objectives

The main objective of this project is to develop a context-to-consequence framework that connects real political news with constituency-level voting outcomes. More specifically, the project aims to:

1. collect and organise political news and constituency-level election data;
2. use LLMs to extract structured context from political news, including affected parties, voter groups, regions, time horizon, expected impact, and rationale;
3. learn consequence-aware news representations using contrastive learning, so that news events with similar voting consequences are closer in the embedding space;
4. combine news representations with constituency-level data to predict voting outcomes, and evaluate the results against baseline methods such as sentiment-based features and raw text embeddings.

Methodology: Context-to-Consequence Pipeline

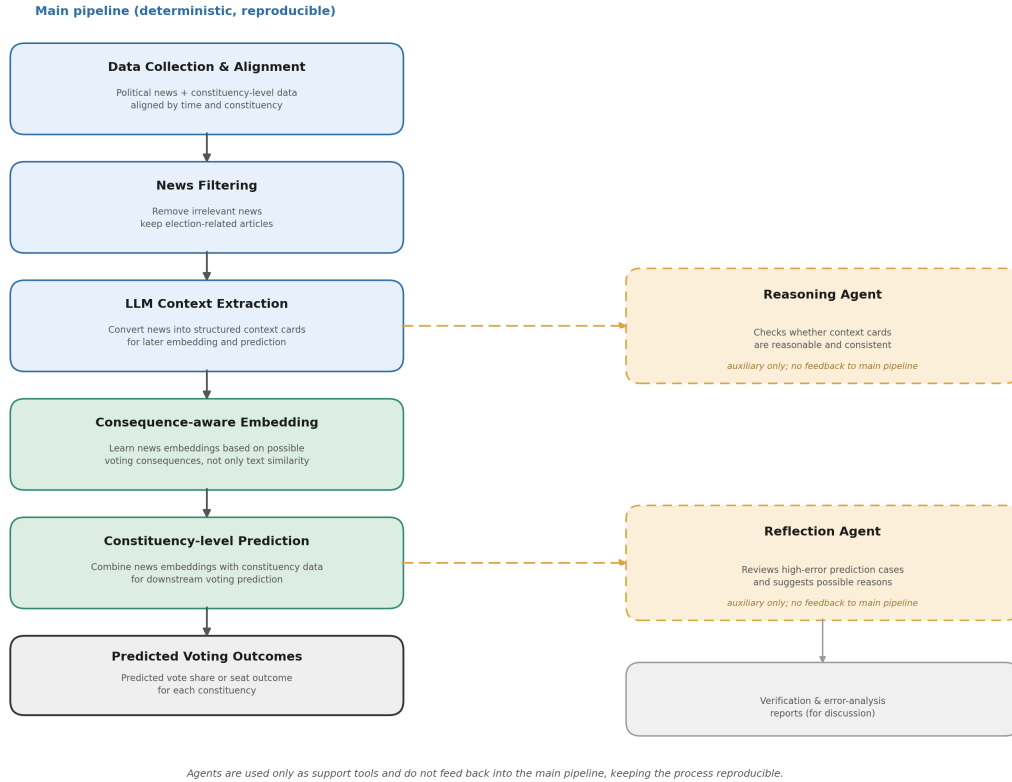


Figure 1: Overview of the context-to-consequence methodology

4 Methodology

4.1 Data Collection and Alignment

The first stage is to collect political news and constituency-level election data. Political news will provide campaign-related context, while election data will provide constituency-level outcomes and background information. At this stage, the two sources will be organised by time, party, region, and constituency where possible.

4.2 News Filtering

According to the research above, Unprocessed political news will not be directly added to the prediction model[7]. Therefore, this project will first filter the collected news before context extraction and prediction.

I plan to use a LLM-based reasoning agents for news filtering [7]. The agent will judge whether each article is relevant to voting outcomes by checking its political and geographical relevance. Irrelevant articles will be excluded from the later prediction pipeline. To reduce data leakage, only news published before a fixed prediction cut-off date will be used.

4.3 LLM-based Context Extraction

After news filtering, this project will use an LLM to convert retained political news into structured context cards for later embedding and prediction. This follows Wang et al., who show that LLM-based agents can filter, classify, and interpret news by producing structured outputs with information such as affected region, reporting time, and rationale [7]. This project will therefore be based on this approach and will be applied to political news and voting outcome prediction.

For this project, each selected news article may be input into LLM along with its publication date, source, and possible geographical information. I will first design a fixed prompt and preliminary output format to guide LLM in extracting information related to the political context, such as news topics, event types, affected parties and so on. To improve consistency, the output will be in a fixed format such as JSON or a table, and basic checks will be performed.

4.4 Consequence-aware Representation Learning

Ordinary text embeddings mainly reflect semantic similarity, but this project I will focus on whether political news may lead to similar voting consequences. For example, news about NHS waiting lists and news about GP appointment access may use different wording, but both could affect healthcare-focused voters and similar types of constituencies. Therefore, this project will transform structured context cards into embeddings and use contrastive learning to organise the embedding space.

I will use two strands of contrastive learning literature in this project. Khosla et al.'s label information method [6] and Gao et al.'s SimCSE [3] method as I noticed before. The aim is to place news with similar possible voting consequences closer together, while separating news with different consequences.

4.5 Downstream Voting Outcome Prediction

After obtaining consequence-aware news representations, the next step will combine them with available constituency-level data for downstream voting outcome prediction. Depending on data availability, these features may include historical election results, constituency background information, and other variables related to voting behaviour. This allows the prediction model to use both traditional constituency-level information and political news context from the election campaign.

This stage will also allow the project to test whether political news context provides additional predictive value. The proposed approach can be compared with simpler baselines.

4.6 Reflection Agent for Error Analysis

After the downstream prediction stage, this project will evaluate prediction performance and consider adding a constrained reflection agent as a support tool for error analysis. Depending on the final prediction target, evaluation may use metrics such as MAE or RMSE for vote share prediction, or accuracy for seat or winning party prediction. The model may also be compared with simpler baselines, such as models using only constituency-level data or simple text-based features.

For high-error cases, the reflection agent may help examine possible reasons, such as missed local news, inaccurate links between news and constituencies, or unsuitable time horizon judgements. Based on this analysis, the project will make improvements to some parts which are perform poorly. If time allows, possible extensions will also be considered, supported by further reading on Related papers.

5 Deliverables

The expected deliverable is a prototype end-to-end pipeline that links political news context to downstream voting outcome prediction. This pipeline will include news filtering, LLM-based context extraction, consequence-aware representation learning, and a prediction stage that combines news representations with available constituency-level data.

If possible, the project may also deliver baseline comparison results, a summary of prediction performance, and error analysis check and improvement supported by a constrained reflection agent. And some extensions if needed.

6 Future Plan

The project is scheduled over ten weeks, from 22 June to the final deadline on 28 August, following the stages described in the Methodology. Figure 2 summarises the weekly plan.

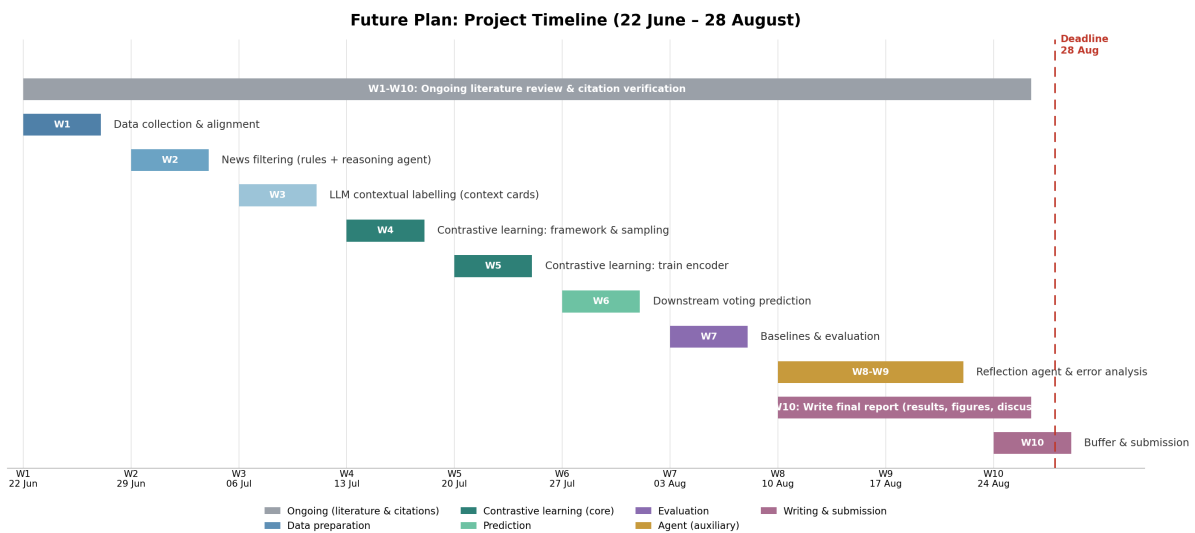


Figure 2: Project timeline (22 June – 28 August).

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